The minimum bisector energy of a planar point set

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Abstract

The *bisector energy* $\mathrm{E}(P)$ of a finite point set $P \subset \mathbb{R}^2$, introduced by Lund, Sheffer, and de Zeeuw, counts ordered quadruples of distinct point pairs sharing a perpendicular bisector. We prove that $\mathrm{E}(P) \geq 2n(n-1)$ for every n-point planar set, with equality if and only if every two distinct point pairs have distinct perpendicular bisectors (*bisector-injective*). The bound is tight: for every finite set G in any Euclidean space with no three collinear points, a single linear projection $T\colon \mathbb{R}^d \to \mathbb{R}^2$ exists such that the image $T(G)$ is bisector-injective, hence attains the floor $\mathrm{E}(T(G)) = 2n(n-1)$. The projected set also has zero rotational energy and its distance multiset is in bijection with the $\pm$-difference classes of G. The proof is verified in Lean 4 against standard Mathlib axioms only (propext, Classical.choice, Quot.sound).

1 Introduction

For a finite set $P \subset \mathbb{R}^2$ with $|P| = n$, define the *bisector energy*

$$\mathrm{E}(P) \;=\; \big|\{(a,b,c,e) \in P^4 : a \neq b,\ c \neq e,\ \beta(a,b) = \beta(c,e)\}\big|,$$

where $\beta(p,q) = \{x \in \mathbb{R}^2 : |x-p| = |x-q|\}$ is the perpendicular bisector of the pair (p,q). This quantity was introduced by Lund, Sheffer, and de Zeeuw [4], who proved the upper bound

$$\mathrm{E}(P) \;=\; O\big(M(n)^{2/5}\, n^{12/5+\varepsilon} + M(n)\, n^2\big)\,,$$

where $M(n)$ is the maximum number of points of P on any line or circle, and applied it to obtain structural results about point sets with few distinct distances.

The present note addresses the complementary question: *how small can* $\mathrm{E}(P)$ *be?*

Every unordered pair $\{a,b\} \subseteq P$ $(a \neq b)$ contributes exactly four ordered quadruples to $\mathrm{E}(P)$ by taking both orderings of each pair: (a,b,a,b), (a,b,b,a), (b,a,a,b), (b,a,b,a). These *trivial quadruples* account for $4\binom{n}{2} = 2n(n-1)$ contributions, so $\mathrm{E}(P) \geq 2n(n-1)$ for every P. A nontrivial contribution arises precisely when two distinct unordered pairs share a perpendicular bisector. We say P is *bisector-injective* if no such coincidence occurs, i.e. the map $\{a,b\} \mapsto \beta(a,b)$ is injective on unordered pairs.

The main theorem shows that the floor $2n(n-1)$ is attained.

Theorem 1 (Near Enemy Theorem for Bisector Energy). *Let G be a finite set in a Euclidean space $\mathbb{R}^d$ with no three collinear points, and let $n = |G|$. There exists a linear map $T\colon \mathbb{R}^d \to \mathbb{R}^2$ such that:*

(a) *T is injective on G;*

(b) $\mathrm{E}(T(G)) = 2n(n-1)$ (bisector-energy floor)*;*

(c) $\mathrm{E}(T(G)) \le \mathrm{E}(P')$ *for every n-point planar set P'* (absolute minimality)*;*

(d) *$T(G)$ has no three collinear points and no four concyclic points* (full general position)*;*

(e) $\mathrm{E}_{\mathrm{rot}}(T(G)) = 0$ (zero rotational energy)*;*

(f) *the distance multiset of $T(G)$ is in bijection with the $\pm$-difference classes of G* (distance transport)*.*

The EFPR construction—the lattice-sphere slice $G = \{x \in [-h,h]^d \cap \mathbb{Z}^d : \|x\|^2 = R\}$ of Erdős, Füredi, Pach, and Ruzsa [1]—satisfies the no-three-collinear hypothesis (a line meets a sphere in at most two points), so the theorem applies. The projected EFPR family is the canonical minimizer.

Notation. We write $\langle \cdot, \cdot \rangle$ for the standard inner product on $\mathbb{R}^d$, $\| \cdot \|$ for the induced norm, and $|S|$ for the cardinality of a finite set S. For a linear map $T\colon \mathbb{R}^d \to \mathbb{R}^2$ we write $T_1, T_2 \in \mathbb{R}^d$ for its two row vectors, so $T(v) = (\langle T_1, v\rangle, \langle T_2, v\rangle)$. The *$\pm$-difference classes* of G are the equivalence classes of $G - G \setminus \{0\} := \{a - b : a, b \in G,\ a \neq b\}$ under $u \sim v$ iff $u = \pm v$.

2 Definitions and the lower bound

Definition 2. *A finite set $P \subset \mathbb{R}^2$ is* bisector-injective *if $\beta(a,b) \neq \beta(c,e)$ for every two distinct unordered pairs $\{a,b\}, \{c,e\} \subseteq P$ with $a \neq b$ and $c \neq e$.*

Lemma 3 (Floor and equality)**.** *For every n-point set $P \subset \mathbb{R}^2$:*

$$\mathrm{E}(P) \ \ge\ 2n(n-1),$$

with equality if and only if P is bisector-injective.

Proof. For each ordered pair (a,b) with $a \neq b$, the quadruples (a,b,a,b) and (a,b,b,a) both lie in the defining set of $\mathrm{E}(P)$ since $\beta(a,b) = \beta(b,a)$. As (a,b) ranges over the $n(n-1)$ ordered pairs with $a \neq b$, these $2n(n-1)$ quadruples are pairwise distinct (the first ordered pair (a,b) and the swap bit together uniquely determine the quadruple), so $\mathrm{E}(P) \ge 2n(n-1)$.

If P is bisector-injective then every quadruple counted by $\mathrm{E}(P)$ has $\beta(a,b) = \beta(c,e)$, hence $\{a,b\} = \{c,e\}$ as unordered pairs, so the quadruple is trivial. Thus $\mathrm{E}(P) = 2n(n-1)$.

Conversely, if P is not bisector-injective there exist distinct unordered pairs $\{a,b\} \neq \{c,e\}$ with $\beta(a,b) = \beta(c,e)$. The four ordered quadruples (a,b,c,e), (a,b,e,c), (b,a,c,e), (b,a,e,c) are all counted by $\mathrm{E}(P)$ and are nontrivial, giving $\mathrm{E}(P) > 2n(n-1)$. $\square$

3 The key criterion: shared bisectors upstairs

The proof of Theorem 1 uses a criterion that translates a shared bisector downstairs into algebraic conditions on the preimages upstairs.

Lemma 4 (Shared-bisector conditions)**.** *Let $p, q, p', q' \in \mathbb{R}^2$. If $\beta(p,q) = \beta(p',q')$ then:*

(i) $q' - p' = t(q-p)$ *for some* $t \in \mathbb{R}$ (parallel differences)*;*

(ii) $\langle p + q - (p' + q'), q - p\rangle = 0$ (midpoint difference orthogonal to direction).

Proof. Two perpendicular bisectors are equal as affine subspaces if and only if they have the same direction and share a point. The direction of $\beta(p,q)$ is the orthogonal complement of $\mathbb{R}(q-p)$; equal directions give (i). Both midpoints $\frac{p+q}{2}$ and $\frac{p'+q'}{2}$ lie on the common bisector, so their difference lies in its direction subspace, giving (ii). □

Lemma 5 (Upstairs line rigidity)**.** *Let $G \subset \mathbb{R}^d$ have no three collinear points, and let $a, b, c, e \in G$ with $a \neq b$ and $c \neq e$. If*

(i) $e - c = t(b-a)$ *for some* $t \in \mathbb{R}$*, and*

(ii) $\langle a + b - (c+e), b - a\rangle = 0$*,*

then $\{a,b\} = \{c,e\}$*.*

Proof. From (i), all four points lie on the affine line $\{a + s(b-a) : s \in \mathbb{R}\}$. Write $c = a + s(b-a)$ and $e = a + (s+t)(b-a)$ for some $s \in \mathbb{R}$. Substituting into (ii):

$$\langle a + b - (c+e), b-a\rangle \;=\; \langle (1-2s-t)(b-a), b-a\rangle \;=\; (1-2s-t)\|b-a\|^2 \;=\; 0.$$

Since $a \neq b$, we get $t = 1 - 2s$, hence $e = a + (1-s)(b-a)$. If $s \notin \{0,1\}$ then a, b, $c = a + s(b-a)$ are three distinct collinear points, contradicting the no-three-collinear hypothesis. Hence $s \in \{0,1\}$: if $s = 0$ then $c = a$ and $e = b$; if $s = 1$ then $c = b$ and $e = a$. In either case $\{c,e\} = \{a,b\}$. □

Remark 6. *The sphere-slice case (where G lies on a sphere) gives a cleaner proof of Lemma 5: chords of a sphere with the same midpoint have equal length (the parallelogram law expresses $\|a-b\|^2$ in terms of $\|a+b\|/2$ and the radius, both fixed by the common midpoint and sphere). Two parallel chords with equal length and the same midpoint then coincide: the midpoint pins the foot of the perpendicular from the center, and equal length pins the two endpoints symmetrically about it. The no-three-collinear hypothesis makes the sphere-slice hypothesis unnecessary for the headline theorem, since a sphere already has no three collinear points.*

4 Generic projections

Definition 7. *A linear map $T\colon \mathbb{R}^d \to \mathbb{R}^2$ is* generic for G *if:*

(inj) $T(a) \neq T(b)$ *for all distinct* $a, b \in G$*;*

(avoid) *for every* $a, b, c, e \in G$ *with* $a \neq b$, $c \neq e$, *and* $\{a, b\} \neq \{c, e\}$*: it is not the case that both* $T(e) - T(c) = t \cdot (T(b) - T(a))$ *for some* $t \in \mathbb{R}$ and $\langle T(a) + T(b) - (T(c) + T(e)), T(b) - T(a)\rangle = 0$*;*

(sep) *for every* $a, b, c, e \in G$ *with* $a - b \neq \pm(c - e)$*:* $\|T(a - b)\|^2 \neq \|T(c - e)\|^2$.

Lemma 8 (Generic projection implies bisector-injectivity)**.** *If T is generic for G then $T(G)$ is bisector-injective.*

Proof. Suppose $T(a), T(b), T(c), T(e) \in T(G)$ satisfy $\beta(T(a), T(b)) = \beta(T(c), T(e))$ with $T(a) \neq T(b)$ and $T(c) \neq T(e)$. By injectivity of T on G, we have $a \neq b$ and $c \neq e$. By Lemma 4, conditions (i) and (ii) hold downstairs. Linearity of T lifts them: $T(e - c) = tT(b - a)$ and $\langle T(a + b - (c + e)), T(b - a)\rangle = 0$. The avoid condition then forces $\{a, b\} = \{c, e\}$, hence $\{T(a), T(b)\} = \{T(c), T(e)\}$. □

Lemma 9 (Existence of a generic projection)**.** *If $G \subset \mathbb{R}^d$ is a finite set with no three collinear points, a generic projection $T\colon \mathbb{R}^d \to \mathbb{R}^2$ for G exists.*

Proof. We view the rows $T_1, T_2 \in \mathbb{R}^d$ as $2d$ free real variables. For each off-pair quadruple (a, b, c, e) (with $a \neq b$, $c \neq e$, $\{a, b\} \neq \{c, e\}$), define two polynomials in (T_1, T_2):

$$D(a, b, c, e) = \langle T_1, b - a\rangle\langle T_2, e - c\rangle - \langle T_2, b - a\rangle\langle T_1, e - c\rangle,$$
$$H(a, b, c, e) = \langle T_1, a + b - (c + e)\rangle\langle T_1, b - a\rangle + \langle T_2, a + b - (c + e)\rangle\langle T_2, b - a\rangle.$$

If both $D = 0$ and $H = 0$ at some T, then condition (avoid) fails, and the proof of Lemma 8 would give $\{a, b\} = \{c, e\}$, contradicting the off-pair assumption. Hence for each off-pair quadruple, $D^2 + H^2$ is a nonzero polynomial in $2d$ variables; its real zero-set has measure zero. Similarly, for each pair of distinct $\pm$-difference classes $[u] \neq [v]$ in $G - G \setminus \{0\}$, the polynomial $\sum_k \langle T_k, u - v\rangle\langle T_k, u + v\rangle$ (encoding condition (sep)) is nonzero: since $u \neq \pm v$, both $u - v \neq 0$ and $u + v \neq 0$, so neither linear factor vanishes identically, and their inner product $\|Tu\|^2 - \|Tv\|^2$ is a nonzero degree-2 polynomial. Likewise the injectivity condition (inj) cuts out a measure-zero set.

The set of $(T_1, T_2) \in \mathbb{R}^{2d}$ satisfying all three conditions is the complement of finitely many measure-zero algebraic sets, hence nonempty (in fact dense) [5]. □

5 Proof of Theorem 1

Proof of Theorem 1. Parts (a)–(c) combine Lemmas 3, 8, and 9: take T generic for G, which exists by Lemma 9. Then T is injective on G (a), $T(G)$ is bisector-injective so

$\mathrm{E}(T(G)) = 2n(n-1)$ by Lemma 3 (b), and $2n(n-1)$ is the global minimum by the lower bound of Lemma 3 (c).

For part (d), one adds two more avoidance conditions to the definition of generic: no collinear triple and no concyclic quadruple in $T(G)$. For collinear triples: $T(a), T(b), T(c)$ are collinear iff $D_{abc}(T) = \langle T_1, b-a\rangle\langle T_2, c-a\rangle - \langle T_2, b-a\rangle\langle T_1, c-a\rangle = 0$. Since a, b, c are not collinear in G, the vectors $b-a$ and $c-a$ are linearly independent, so D_{abc} is a nonzero polynomial. For concyclic quadruples: the 4×4 circle determinant is likewise a nonzero polynomial in (T_1, T_2), provided no three of the four upstairs points are collinear—a condition guaranteed by the no-three-collinear hypothesis on G. The same measure-zero argument applies.

For part (e), write $\mathrm{E}_{\mathrm{rot}}(T(G))$ as the count of ordered quadruples $(p, q, r, s) \in T(G)^4$ with $p \neq q$, $r \neq s$, $|p-q| = |r-s|$, and $p-q \notin \{r-s, -(r-s)\}$. The condition $|p-q| = |r-s|$ is equivalent to $\|T(a-b)\| = \|T(c-e)\|$, which expands as $\|T(a-b)\|^2 - \|T(c-e)\|^2 = \sum_k \langle T_k, (a-b)-(c-e)\rangle\langle T_k, (a-b)+(c-e)\rangle = 0$. Condition (sep) forces $a - b = \pm(c-e)$, i.e. $p - q = \pm(r-s)$, so the proper-rotation channel is empty and $\mathrm{E}_{\mathrm{rot}}(T(G)) = 0$.

For part (f), the same separation argument shows that the map $T(G) - T(G) \setminus \{0\} \to G - G \setminus \{0\}$ preserves $\pm$-classes, giving a bijection between the sets of distinct distances. $\square$

6 Corollaries

Corollary 10 (Sphere-slice form)**.** *Let G be a finite subset of a sphere in $\mathbb{R}^d$, $|G| = n$. Then a linear map $T\colon \mathbb{R}^d \to \mathbb{R}^2$ exists satisfying* (a)–(f) *of Theorem 1.*

Proof. A line meets a sphere in at most two points, so G has no three collinear points. Apply Theorem 1. $\square$

Corollary 11 (EFPR minimizer)**.** *The EFPR lattice-sphere-slice family $G = \{x \in [-h, h]^d \cap \mathbb{Z}^d : \|x\|^2 = R\}$ [1] admits a planar projection that attains the bisector-energy floor. In particular, the minimum bisector energy of an n-point planar set is exactly $2n(n-1)$.*

Proof. Apply Corollary 10 to the EFPR lattice points on a sphere. $\square$

Remark 12 (Relation to LSdZ)**.** *Lund, Sheffer, and de Zeeuw [4] remark in passing that $\mathrm{E}(P) = \Omega(n^2)$, but do not extract the constant $2n(n-1)$, state the floor, or characterize equality. The minimization direction (Lemma 3), the equality condition (bisector-injectivity), and the identification of the EFPR projection as a minimizer (Corollary 10) appear to be new.*

Remark 13 (Rotation energy)**.** *Part (e) says the near-enemy projection has zero rotational energy in the sense of the Elekes–Sharir / Guth–Katz framework [2, 3]: every congruent quadruple (a, b, c, e) in $T(G)$ is realized by a translation ($a - b = c - e$) or half-turn ($a - b = -(c-e)$), never by a proper rotation. This is the strongest possible statement: a set with any two congruent pairs at a proper-rotation angle apart contributes to the rotation channel.*

7 Verification

All results in this paper are verified in Lean 4 using Mathlib. The Lean source is `NearEnemyTheorem.lean` in the `NearEnemy` namespace of the `lean-formalizations` repository. The headline Lean theorem corresponding to Theorem 1 is

`nearEnemy_noThreeCollinear_exists_bisectorEnergy_minimal`
`_image_generalPosition_distanceTransport`.

Running `#print axioms` on this theorem confirms that the proof depends only on `propext`, `Classical.choice`, and `Quot.sound` — the standard Lean/Mathlib axioms. No `sorry` appears anywhere in the proof chain.

References

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[2] G. Elekes and M. Sharir, *Incidences in three dimensions and distinct distances in the plane*, Combin. Probab. Comput. **20** (2011), no. 4, 571–608.

[3] L. Guth and N. H. Katz, *On the Erdős distinct distances problem in the plane*, Ann. of Math. (2) **181** (2015), no. 1, 155–190.

[4] B. Lund, A. Sheffer, and F. de Zeeuw, *Bisector energy and few distinct distances*, Discrete Comput. Geom. **56** (2016), no. 2, 337–356. arXiv:1411.6868.

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